
TITAGARH
RAIL SYSTEMS LTD.
#MOBILITYFORBILLIONS

RDSO PRESENTATION · 2026

Digitization of **Foundry & Wagon** **Plant Operations**

Quality & Safety Digitization Initiative — building a fully traceable, digitally controlled, intelligent manufacturing ecosystem.

PRESENTED TO

RDSO — Research Designs & Standards Organisation

PRESENTED BY

Titagarh Rail Systems Ltd. | Powered by MetalCloud

VISION: Create a fully traceable, digitally controlled, intelligent manufacturing ecosystem aligned with future-ready railway manufacturing standards.

KEY FOCUS AREAS

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Quality Traceability

- 100% casting traceability
- Digital quality gates
- Automated inspection workflows
- Serial-number-based tracking
- Audit-ready manufacturing

Safety Digitization

- AI CCTV monitoring
- PPE compliance detection
- Safety violation alerts
- Real-time monitoring
- Zero uncontrolled violations

INITIATIVE OVERVIEW

Strategic Objectives & Expected Outcomes

Two core pillars driving operational transformation at Titagarh Rail Systems

About Titagarh Rail Systems Ltd.

India's leading mobility solutions provider with operations across India and Italy — delivering semi high-speed trains, urban metros, passenger coaches, propulsion systems, and specialised wagons. Guided by innovation and operational excellence with a vision for sustainable, future-ready manufacturing.



Digital Transformation (MetalCloud)

Real-time visibility beyond melting — covering Procurement, Moulding, Fettling, Machining, Assembly, Quality, Dispatch & Safety via IoT sensors, AI Vision, and Kiosk/App tracking.



Process Excellence & Single Truth

Single source of truth across Quality, Production & Safety. Maker–checker workflows enforced. Heat-level and serial-level traceability for every casting.

PROJECTED OUTCOMES

10–15%	20–25%	80%	100%
Higher Throughput	Fewer Delivery Delays	Fewer Post-Dispatch Claims	Safety Compliance

LIVE FROM 1 FEBRUARY 2026

Current Digitization Status — What Is Live Today

19 Devices Deployed · Systems Live | **Phase 1 · 100% Complete** | **Phase 2 Automation · 50% Complete** | **app.nowpurchase.com**

Melting Digitization

LIVE

- All melting log sheets digitised
- Heat tracking active & live
- Power & pyrometer integration
- Tap-to-tap time tracking

Production Tracking

LIVE

- Core making / Moulding / Pouring
- Shakeout / Fettling / Machining
- Assembly / Dispatch workflows
- 18 tablets on shop floor

Digital Traceability

LIVE

- Serial-number-based tracking
- Complete casting lifecycle visible
- Real-time WIP at every stage
- Stage-wise progress tracking

Quality Lab Integration

IN PROGRESS

- Spectrometer ✓ Live
- Hardness · Microstructure ✓ Live
- Tensile / UTM ✓ Live
- Heat treatment integration (ongoing)

Dashboards & Analytics

LIVE

- Dispatch visibility dashboard
- Rejection trends & analysis
- Live WIP & aging reports
- Power BI-ready Excel exports

IoT & Sensor Integration

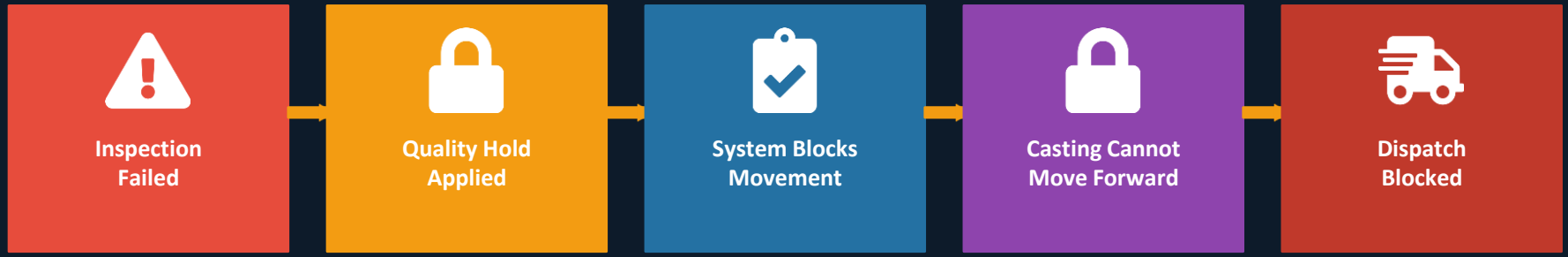
IN PROGRESS

- Energy meter ✓ Live
- Pouring time via AI CCTV ✓ Live
- Heat treatment params (ongoing)
- Sand consumption (pending)

QUALITY ENFORCEMENT

Digital Quality Gate — Zero Bypass Architecture

No casting can leave the factory unless all digital quality gates are cleared



- Quality inspection PENDING or FAILED → system BLOCKS progression to next stage
- Maker–checker mechanism enforced — every quality input verified before clearance
- Next stage clearly shows "Quality Pending" — zero process bypass possible
- Serial number unique & non-duplicable — traceability cannot be forged or skipped

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End-to-End Digitised Production Flow — 13 Sections

Every production stage produces structured digital data — no paper, no parallel logs

01	02	03	04	05	06	07	08	09	10	11	12	13
Inward Inspection	Moulding & Pouring	Melting	Shake Out	Shot Blasting	Heat Treatment	Fettling	Machining	Assembly	Painting	Quality Test	AI Visual Inspection	Dispatch
4 Auto	4 Auto	6 Auto	4 Auto	4 Auto	5 Auto	4 Auto	4 Auto	4 Auto	4 Auto	6 Auto	5 Auto	5 Auto
3 Semi	2 Semi	2 Semi	2 Semi	2 Semi	2 Semi	2 Semi	2 Semi	3 Semi	3 Semi	5 Semi	3 Semi	2 Semi

15	69	37	13	9
Total Initiatives	Automated Data Points	Semi-Automated Points	Production Stages	Quality Impact Areas

AI-POWERED INTELLIGENCE

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Quality & Safety Intelligence Layer

Automated detection, real-time alerts, and closed-loop accountability

⚙ QUALITY DIGITIZATION

- Spectrometer deviation auto-alerts
- Hardness & tensile auto-capture
- Automated lab instrument integration
- AI defect identification (Phase 2)
- Real-time quality deviation alerts
- End-to-end serial traceability
- Maker–checker workflow enforced
- Audit-ready documentation

TRACEABILITY DEMO — Serial #222869

Melt → Pour → Shakeout → HT → Inspection → Dispatch

🛡 SAFETY DIGITIZATION · AI · NETTO CLOUD

- AI CCTV monitoring 24 × 7
- Helmet detection & alerts
- Safety shoe compliance
- Unsafe zone entry detection
- Violation logging & tracking
- Real-time WhatsApp / Email alerts
- End-of-day violation summary
- Zone-wise surveillance coverage

LIVE SAFETY ZONES — COMMAND CENTER

Melting | Moulding | Fettling | Pouring | Heat Tx | Dispatch

From manual monitoring to intelligent real-time factory governance — powered by AI and connected IoT.

DIGITAL MANUFACTURING

Digitization Roadmap

A structured journey from digitally controlled foundry to AI-led, self-governing manufacturing ecosystem

Nov 2025 — FEB 2026 | FEB 2026 — JUL 2026 | JUL 2026 — DEC 2026

Phase 1 · FOUNDATION
COMPLETED ✓

100%

- 19 tablets & mobiles deployed
- All log sheets digitised
- Core / Moulding / Melting covered
- Serial traceability live
- Dispatch digitization complete
- Quality gates implemented
- 7 Milestones closed (M1–M7)

Phase 2 · AUTOMATION
50% COMPLETE

50%

- Spectrometer / Hardness / UTM live
- Pouring temperature captured
- Pouring time via AI CCTV live
- Energy meter integrated
- Heat treatment (in progress)
- Sand consumption (pending)
- Vision-based inspection

Phase 3 · INTELLIGENT FACTORY
PLANNED

12%

- Full machine connectivity
- Predictive quality analytics
- AI defect recognition
- Factory command centre
- Real-time RDSO reporting
- Zero-paper autonomous ops
- SAP full integration

VALUE DELIVERED

Operational Outcomes — What Digitization Delivers

PRODUCTIVITY

- 20–25% throughput improvement
- 15–20% delivery-performance gain
- Real-time MIS replacing manual reporting
- Automated shift/section dashboards
- WIP control & bottleneck visibility

QUALITY

- 100% castings through digital quality gate
- 80% reduction in warranty/post-dispatch claims
- Heat-wise, batch-wise, casting-wise traceability
- Auto-generated RDSO-compliant DM reports
- AI/Vision-based inspection (Phase 2)

COST

- Reduction in rework, rejection & failure costs
- Improved yield through tighter process control
- Lower losses from data-backed decisions
- Better energy/utility utilisation via IoT metering
- Automated contractor payments via WIP reconciliation

SAFETY

- 100% PPE compliance in monitored zones
- AI helmet & safety-shoe detection
- Real-time violation alerts (WhatsApp/Email)
- Plant-wide safety monitoring (Phase 2 expansion)
- End-of-day violation summaries

PHASE 2 · IN PROGRESS

Automation, Intelligence & Adoption — Target 30 May 2026

Building on a live, proven digital foundation — 50% complete

Phase 2 Overall: 50% Complete | Target: 30 May 2026

LIVE

Feedback & Continuous Improvement

Workflow refinement based on Phase-1 feedback. Making the system more intuitive, faster, and aligned with ground-level usage.

LIVE

Continuous Hand-Holding & Adoption Support

Dedicated on-ground & remote support since 1 Feb. Key enabler for sustained adoption — not just system availability.

IN PROGRESS

Automated Data Capture & IoT Digitisation

IoT engineer onboarded. Energy meter ✓ Pouring temp ✓ AI CCTV time ✓ Heat treatment (in progress) Sand consumption (pending).

IN PROGRESS

Vision-Based Inspection

Surface roughness detection. Visual defect identification. Step toward AI-assisted inspection and consistency enhancement.

IN PROGRESS

Custom MIS & Dream Dashboard

TRSL leadership-defined KPI consolidation. Standard MIS already live — Phase-2 focuses on custom configuration.

IN PROGRESS

Automated Deviation Alerts & Closed-Loop Action

Alerts via System / WhatsApp / Email. Every alert requires an action. Closure timelines visible on dashboards.

PROJECT STATUS

Implementation Progress by Section — Live Overview

Overall completion: 78% · All Phase-1 sections completed · Phase-2 automation underway

78% OVERALL COMPLETE

Inward Inspection	100%
Pattern / Moulding	100%
Core Making	100%
Melting	100%
Pouring	100%
Shakeout	100%
Fettling	100%
Shot Blasting	100%
Heat Treatment	100%
Final Inspection	25%
Load Test	20%
Assembly	15%
Dispatch	10%

Phase 1: System Development 100% Complete · Phase 2: Equipment Integration & Automation — 10% (Target 15 May 2026) · Live: app.nowpurchase.com

KEY OUTCOME

100% Traceable. 100% Accountable.

Casting Lifecycle — Every Stage Digitally Captured

01	Raw Material Inward	Digital capture	Timestamp	Quality status	Traceability
02	Melting	Digital capture	Timestamp	Quality status	Traceability
03	Moulding & Pouring	Digital capture	Timestamp	Quality status	Traceability
04	Heat Treatment	Digital capture	Timestamp	Quality status	Traceability
05	Inspection & Quality Gate	Digital capture	Timestamp	Quality status	Traceability
06	Assembly	Digital capture	Timestamp	Quality status	Traceability
07	Painting	Digital capture	Timestamp	Quality status	Traceability
08	Dispatch	Digital capture	Timestamp	Quality status	Traceability

19+
Devices Deployed

100%
Digital Traceability

0
Uncontrolled Stage Bypass

AI
Safety Monitoring

LIVE
Operational Dashboards

✓
Full Audit Readiness

"A casting cannot proceed unless digitally approved at every quality checkpoint."

Titagarh Rail Systems is building a future-ready, digitally governed manufacturing ecosystem aligned with the next generation of railway manufacturing standards.

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